

~~date: 07/10/2025~~

date: 09/10/2025

TASK -1 - CRUD OPERATIONS IN GRAPH DATABASES

Aim:

To perform CRUD Operations like creating, inserting, querying, finding, deleting Operations on graph spaces.

• create Node with Properties

Properties are the key-value pairs using which a node stores data. You can create a node with properties using the CREATE clause. You need to specify these properties separated by commas within the flower braces "{}".

Syntax

Following is the syntax to create a node with properties
CREATE (node:label {key1:value, key2:value,})

Renaming the created Node

To verify the creation of the node, type and execute the following query in the dollar prompt, MATCH (n) RETURN n
~~Match~~

creating Relationships

We can create a relationship using the CREATE clause. We will specify relationship within the square braces "[]" depending on the direction the relationship it is placed between hyphen "-" and arrow "→" as shown in the following syntax.

Syntax:

Following is the syntax to create a relationship using the CREATE clause.

CREATE (node1) -[:RelationshipType]-(node2)

Creating a Relationship Between the Existing nodes

You can also create a relationship between the existing nodes the MATCH clause.

Syntax:

Following is the Syntax to create a relationship using the MATCH clause.

```
MATCH (a:LabelOfNode1),(b:LabelOfNode)
WHERE a.name = "name of node 1" AND b.name = "name of node 2"
CREATE (a)-[:Relationship]->(b)
RETURN a,b
```

Deleting a Particular node

To delete a particular node, you need to specify the details of the node in the place of "n" in the above query.

Syntax:

Following is the Syntax to delete a particular node from Neo4j using the DELETE clause.

```
MATCH (node:label {properties.....})
DETACH DELETE node.
```

Create a graph database for student course registration, create student and dept node and insert values of properties

```
create (n:student {sid: "VTU14001", Sname: "Dharsana", deptname: "EEE"})
```

OUTPUT

Added 1 label, created 1 node, set 3 properties, completed after 232ms

```
create (n:student {sid: "VTU14001", Sname: "Dharsana", deptname: "EEE"})
```

PUT:

Added 1 label, created 1 node, set 3 properties, completed after 16ms.

```
create (n:student {sid: "VTU14002",
```

```
name: "vijay",
```

```
deptname: "CSE"
```

07807

Add label, Created 1 node, set 3 properties & completed after
2 ms.

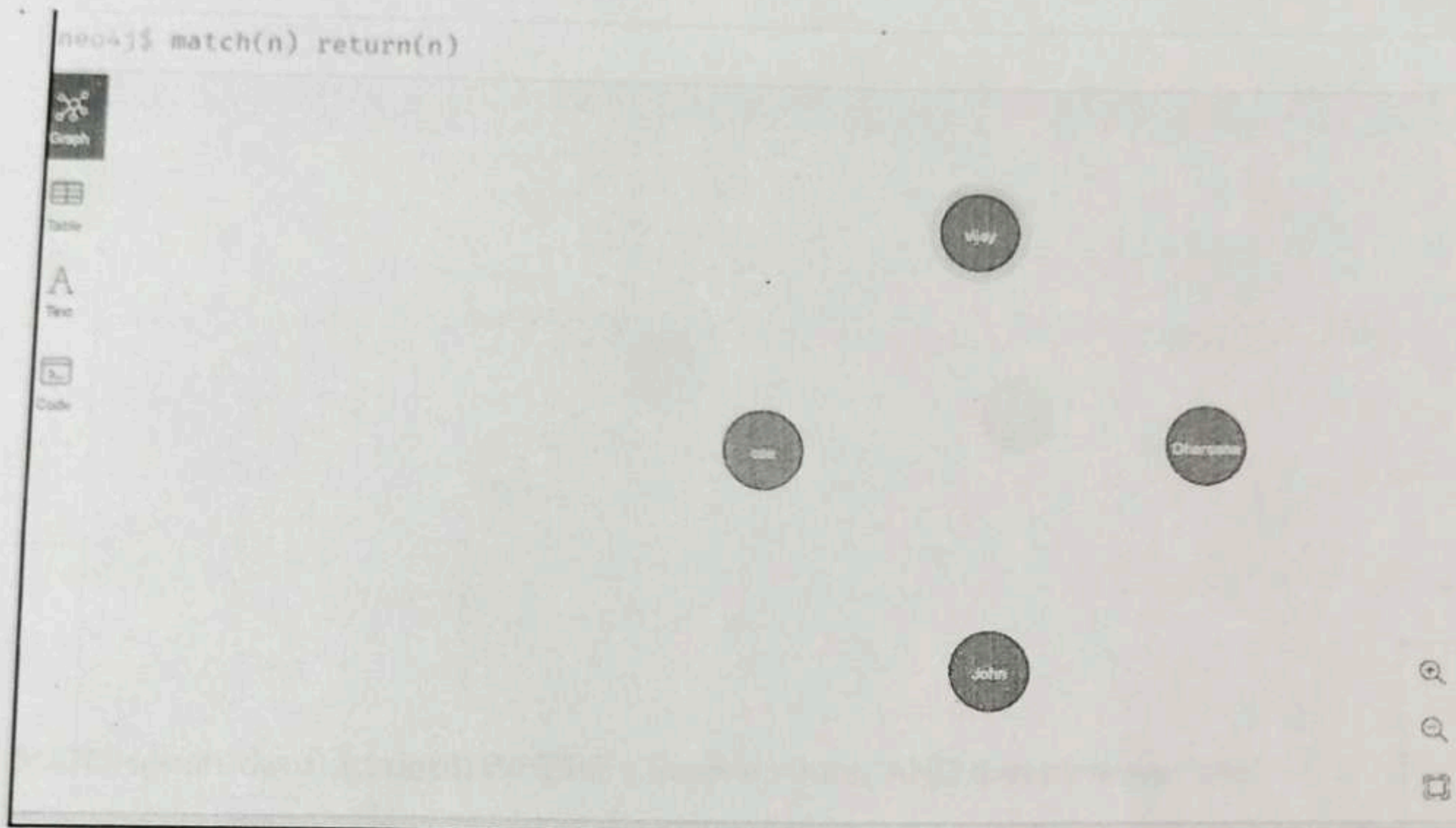
Created (n:dept&deptname:



ASK-11:- CRUD OPERATIONS IN GRAPH DATABASES

Select all the nodes in your database using match command.

```
match(n) return(n)
```



```
match(n:student) return(n)
```

```
neo4j$ match(n:student) return(n)
```



a) Create relationship between student and cse .

```
MATCH(s:student),(d:dept) WHERE s.Sname='vijay' AND d.deptname='cse'
```

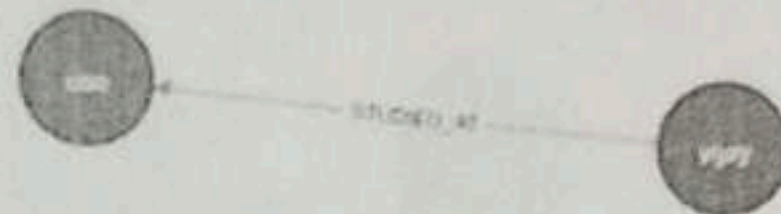
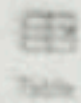
```
CREATE(s)-[st:STUDIED_AT]->(d)
```

```
return s,d
```

```

1 MATCH(s:student),(d:dept) WHERE s.Sname = 'vijay' AND d.deptname = 'cse'
2 CREATE(s)-[st:STUDIED_AT]->(d)
3 return s,d

```



```

MATCH(s:student),(d:dept) WHERE s.Sname = 'John' AND d.deptname = 'cse'

```

```

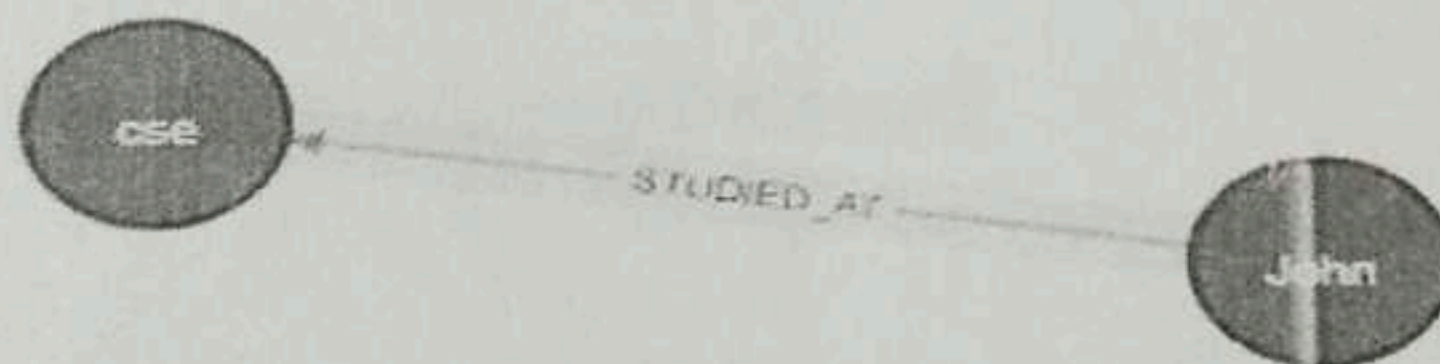
CREATE(s)-[st:STUDIED_AT]->(d)

```

```

return s,d

```



OUTPUT
Addel
12 ms
created

match(n) return(n)

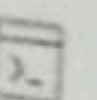
neo4j\$ match(n) return(n)



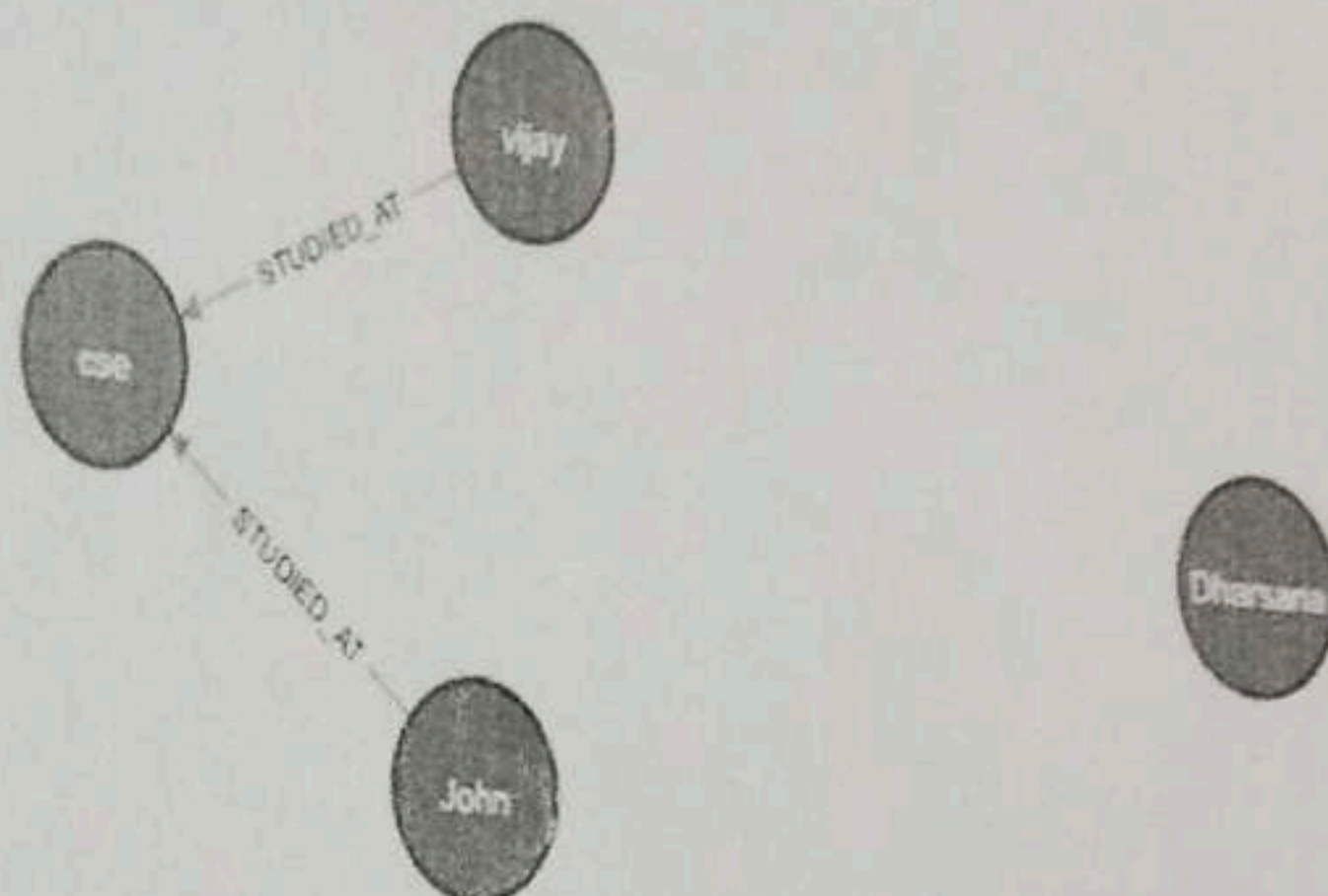
Table



Text



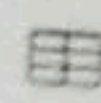
Code



b) Delete a node from student

match(n:student{Sname:'Dharsana'}) Delete(n)

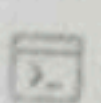
neo4j\$ match(n) return(n)



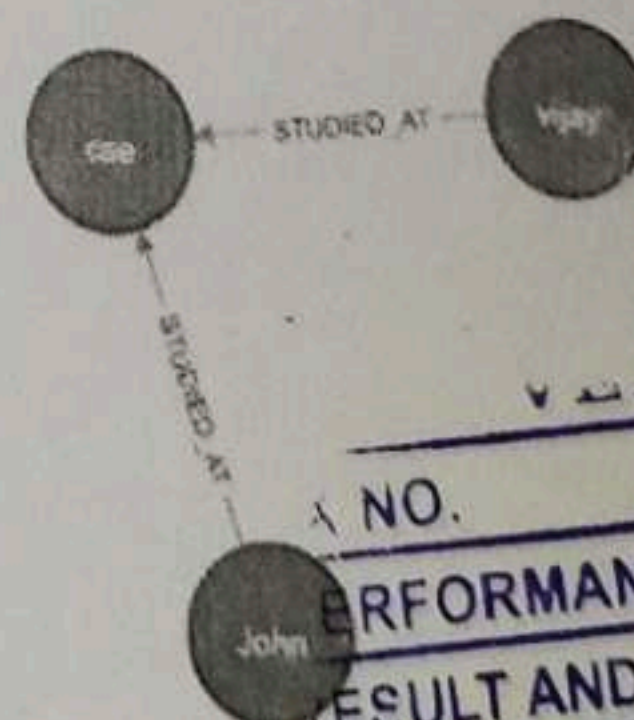
Table



Text



Code



RESULT:- Thus, the implementation of CRUD operations in Graph databases was successfully completed

MARKS	
NO.	
PERFORMANCE (5)	
RESULT AND ANALYSIS (5)	
IVA VOCE (5)	
RECORD (5)	
TOTAL (20)	
SIGN WITH DATE	